

Mark schemes

Q1.

(a) $v = 0 + 0.8 \times 3$

Full method for M1

M1

$v = 2.4$

A1

2

(b)



1st line

2nd line

labels: 3, 7, 2.4

(2.4 may be implied by position between 2 and 3)

B1B1B1

3

(c) $s_1 = \frac{1}{2} \times 3 \times 2.4$ **or** $s_1 = 0 + \frac{1}{2} \times 0.8 \times 3^2$

M1

$s_2 = 9.6$

B1

$s = 13.2$

A1F

average speed = $\frac{13.2}{7}$

$= 1.89 \text{ ms}^{-1}$

A1F

5

Alternative

$$s = \frac{1}{2} (7 + 4) \times 2.4$$

M1A1

$$= 13.2$$

A1F

Allow candidate's total dist. Found in (c)
ft total distance found in (c)

[10]

Q2.

(a) $x = 7t, y = 7t - \frac{1}{2} \times 9.8t^2$ $\left(\begin{matrix} 7t \\ 7t - 4.9t^2 \end{matrix} \right)$

B1 horiz

M1 use of $s = ut + \frac{1}{2}at^2$

M1A1 for vert, accept 10 or g

B1M1A1

3

(b) $t = \frac{x}{7}$

B1

$$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$$

M1A1

$$y = x - 0.1x^2$$

A1

4

Alternative (full verification method)

$x = 7t$ used B1

subs into $y = x - 0.1x^2$

M1A1

$$y = 7t - 4.9t^2 \text{ as in (a)}$$

A1

Alternative: (use equation of path)

$$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$$

45° used

			B1	
	<i>subs</i>			
			M1A1	
	<i>simplify</i>			
			A1	
(c)	$x = 4, \quad y = 4 - 0.1 \times 4^2$			
	$\text{or } t = \frac{4}{7} (0.571(43))$			
			M1	
	$y = 2.4$			
			A1	
	above wire as $2.4 > 2$			
	<i>ft slip in y provided y positive</i>			
			A1F	3
(d)	$y = 0$			
	$x - 0.1x^2 = 0$			
			M1	
	$x(1 - 0.1x) = 0$			
			m1 A1	
	$x = 10$			
			A1F	4
	Alternative: (use of coordinate equations)			
	$y = 0, \quad 7t - 4.9t^2 = 0$			
			M1	
	$\Rightarrow t = \frac{10}{7} \quad (1.43)$			
			m1A1	
	$x = 7t \quad = 10$			
			A1F	
	(Accept 10.01 if $t = 1.43$ used)			
	Alternative: (use of symmetry)			
	$v = 0, \quad 0 = 7 - 9.8t$			

M1

$$t = \frac{5}{7}$$

A1

$$x = 7 \times t \times 2$$

m1

$$x = 10 \text{ (acc 9.996)}$$

A1F

Alternative: (use of equation for Range)

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

Correct formula used M1, 45° used m1

Correct subs A1,

result A1F, AWRT 10m, ft one slip.

(e) $\text{Speed}^2 = 7^2 + 7^2$

M1

$$= 9.90 \text{ ms}^{-1}$$

Accept 9.9, $7\sqrt{2}$,
or 9.91 if $t = 1.43$ used in downward motion

A1

2

[16]

Q3.

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(a) $v^2 = 0 + 2 \times 9.8 \times 10$

Write 'wrong g' at first use of 10 or
9.81 and award A0.

full method for v; $t = 1.42857$ if used

M1

$$v = 14 \text{ ms}^{-1}$$

$t = 1.43$ gives $v = 14.0$, $g = 10$ gives 14.1 (A0)

-9.8 used $\Rightarrow$ M1 A0

A1

2

- (b) Air or wind resistance forces ignored, stone modelled
as a particle, acceleration constant

Not air friction, no external effect

B1

1

[3]

Q4.

(a) (i) $a = \frac{(5U - 3U)}{(5 - 2)}$
full method for a

M1

$a = \frac{2U}{3}$
Accept 0.67U

A1

2

(ii) $s = \frac{1}{2}(5U + 3U) \times 3$
full method for s or attempt at area of whole shape

M1

$s = 12U$
ft incorrect a used

A1F

2

(b) $20 = \frac{1}{2}(5U + 0) \times 4$
*full method, use of equation or area of triangle
 (a = -1.25 U if used)*

M1A1

$U = 2$

ft slip in line above e.g. miscopy

A1F

3

Alternative: average speed = 5

M1

equation for U

A1

$U = 2$

A1

[7]

Q5.

(a) $0 = 21 \times 0.7 - 9.8t$

*Write 'wrong g' at first use of 100 or 9.81 and award A0.
complete method with attempt at
component; accept $\alpha = 44.4(3)$; accept $\pm g$*

M1 A1

$t = 1.5$

*If $\alpha = 44.4$ used, or 2 stage method used,
accept values of t which round to 1.5;
($g = 10$ gives 1.47)*

A1

3

(b) (i) $S = 21 \times 0.7 \times 1.5 - \frac{1}{2} \times 9.8 \times 1.5^2$
*complete method with attempt at
component; accept c's time and $\pm g$*

M1

or $0 = (21 \times 0.7)^2 - 2 \times 9.8 \times S$
all correct, (accept c's time) including $-g$ and $\sin \alpha$

A1

$S = 11.025$

Accept c's time; ft one slip e.g. $+g$ or $\cos \alpha$

A1F

Distance above tree = $11.025 - 8 \approx 3\text{m}$

ft if $d > 0$,

$g = 10$ gives $S = 10.8$ and $d = 2.8$ (accept)

A1F

4

(ii) $S = 8$

$8 = 21 \times 0.7t - \frac{1}{2} \times 9.8 \times t^2$

full method for t or v with attempt at component ;

accept $\pm g$

all correct, including $-g$ and $\sin \alpha$,

but condone repeated error in $-g$ and $\sin \alpha$ from (i)

M1 A1

$$4.9t^2 - 14.7t + 8 = 0$$

$$t = \frac{14.7 \pm \sqrt{14.7^2 - 4 \times 4.9 \times 8}}{2 \times 4.9}$$

To find t by solving equation or using $v (=7.7)$

in $v = u + at$

Substituting; ft

m1 A1F

$$t = 0.71 \text{ (0.7143)}$$

CAO

A1

5

(iii) time = $2.29 - 0.71$

Subtraction of two positive values of t ,
or use of symmetry

M1

or $2(1.5 - 0.71)$

Allow values between 2.28, 2.29 and 0.71, 0.72

$$= 1.57 \text{ sec}$$

Accept 1.56 – 1.58 ($g = 10$ gives $t = 1.5$)

A1F

2

Alternative: If $\cos 44.4$ used:

(b)(i) $S = 11.475$, $d = 3.475$

(ii) $t = 0.688$ (2.373)

(iii) $t = 1.69$

If no component:

(a) M0;

(b)(i) SC: $d = 12.5$

M1 A1 for complete method

(ii) SC: $t = 0.423$

M1 A1 for complete method

(iii) $t = 3.42$

M1A1

[14]

Labelling

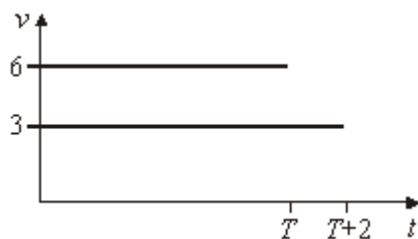
B1

3

[12]

Q7.

(a)



horizontal line below other

B1

(0, 3 to (T + 2, 3)

B1

2

(b) (i) $x + 20$

B1

1

(ii) $x + 20 = 6T$

B1

1

(iii) $3(T + 2) = x$

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2

(iv) $6T - 3T - 6 = 20$

M1

$$3T = 26 \quad T = \frac{26}{3}$$

ft if T positive

A1F

$$\frac{26}{3} \times 6 = x + 20$$

M1

$$x = 32$$

ft if x positive

A1F

4

[10]

Q8.

(a) Vertical: $0 = 4Ut - \frac{1}{2}gt^2$
Accept use of 9.8

M1A1

$$t = \frac{8U}{g}$$

A1

3

(b) horizontal: $OH = 3U \times \frac{8U}{g}$
Use of $v \times t$

M1

$$OH = \frac{24U^2}{g}$$

A1

2

(c) $\text{speed}^2 = (3U)^2 + (4U)^2$

M1

$$\text{speed} = 5U$$

A1

2

(d) $(\sqrt{18}U)^2 = (3U)^2 + v^2$

M!A1

$$v = (\pm)3U$$

A1

$$\pm 3U = 4U - gt$$

Or attempt to solve quadratic in t

M1

$$t_1 = \frac{U}{g} \qquad t_2 = \frac{7U}{g}$$

A1A1

6

[13]

Q9.

(a) (i) $80 = \frac{1}{2} (0 + 20) t$

Use of constant acceleration equation to form an equation for t.

M1

$t = 8 \text{ seconds}$

Correct time

A1

2

(ii) $20^2 = 0^2 + 2 \times a \times 80$

Use of constant acceleration equation to form an equation for a.

M1

$$a = \frac{20^2}{160} = 2.5 \text{ ms}^{-2}$$

ag *Correct acceleration from correct working.*

A1

2

(b) (i) $F = 1200 \times 2.5 = 3000 \text{ N}$

Use of $F = ma$

M1

Correct force.

A1

2

(ii) $F - 400 = 3000$

Three term equation of motion.

M1

$$F = 3400 \text{ N}$$

Correct F .

A1

2

[8]

Q10.

(a) (i) $9 = 18 + 6a$

Use of constant acceleration equation to form an equation for a .

M1

$$a = -\frac{9}{6} = -1.5 \text{ ms}^{-2}$$

Correct equation

A1

ag Correct acceleration from correct working.

A1

3

(ii) $s = 18 \times 6 + \frac{1}{2} \times (-1.5) \times 6^2$

Use of constant acceleration equation to find s .

M1

$$= 81 \text{ m}$$

Correct distance

A1

2

(b) (i) $15 = 9 + 1.2t$

Use of constant acceleration equation to form an equation for t .

M1

$$t = \frac{15 - 9}{1.2} = 5 \text{ s}$$

Correct time

A1

2

(ii) $s = \frac{1}{2} (9 + 15) \times 5$

Use of constant acceleration equation to find s.

M1

Correct expression.

A1

$= 60 \text{ m}$

Correct distance.

A1

3

(c) $81 + 60 = 141 \text{ m}$

Correct sum of distances from (a)(ii) and (b)(ii)

B1

1

[11]

Q11.

(a) $-16\mathbf{i} + 16\mathbf{j} = \frac{1}{2} \mathbf{a} \times 4^2$

Use of vector constant acceleration equation

M1

Correct equation

A1

$\mathbf{a} = \frac{-16\mathbf{i} + 16\mathbf{j}}{8} = -2\mathbf{i} + 2\mathbf{j}$

Solving for a

M1

ag Correct a

A1

4

(b) $\mathbf{v} = 5(-2\mathbf{i} + 2\mathbf{j}) = -10\mathbf{i} + 10\mathbf{j}$

Expression for v at t = 5

M1

Correct expression

A1

$$v = \sqrt{10^2 + 10^2} = 14.1 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Finding magnitude

M1

Correct speed

A1

4

(c) $20\mathbf{i} - 10\mathbf{j} + \mathbf{Q} = 3(-2\mathbf{i} + 2\mathbf{j})$

Application of Newton's second law to form equation

M1

Correct equation

A1

$$\mathbf{Q} = (-20 - 6)\mathbf{i} + (10 + 6)\mathbf{j} = -26\mathbf{i} + 16\mathbf{j}$$

Solving for Q

M1

Correct Q

A1

4

[12]

Q12.

(a) $25\cos 17^\circ t = 12$

Equation to find time of flight

Correct equation

M1 A1

$$t = \frac{12}{25\cos 17^\circ} = 0.502 \text{ s}$$

ag *Correct time from correct working*

A1

3

(b) $h = 25 \sin 17^\circ \times 0.502 - 4.9 \times 0.502^2$

Substituting $t = 0.502$ into an expression for the height of the ball

M1

Correct expression

A1

$= 2.43 \text{ m}$

Correct height

A1

3

(c) $v_x = 25\cos 17^\circ$

Finding horizontal and vertical components of velocity

M1

$v_y = 25\sin 17^\circ - 9.8 \times 0.502 = 2.390$

Horizontal component correct

A1

Vertical component correct

A1

$\tan \alpha = \frac{25\cos 17^\circ}{2.390}$

Use of tan to find angle

M1

$\alpha = 84.3^\circ$

Correct angle

A1

5

(d) Ball is a particle

First assumption

B1

No air resistance

Second assumption

B1

2

[13]

Q13.

(a) $4^2 = 20^2 + 2 \times 48a$

Use of constant acceleration equation to find a

M1

$$a = \frac{16 - 400}{96} = -4 \text{ ms}^{-2}$$

ag Correct acceleration from correct working

A1

2

(b) $0^2 = 4^2 + 2 \times (-4)s$

use of constant acceleration equation to form equation for s

M1

$$s = \frac{16}{8} = 2 \text{ m}$$

Correct equation

A1

Correct s

A1

3

(c) $0 = 20 - 4t$

Use of constant acceleration equation to form equation for t

M1

$$t = 5$$

Correct t

A1

2

(d) $-F - 200 = 1100 \times (-4)$

Three term equation of motion

M1

Correct equation

A1

$$F = 4200$$

Correct F

A1

3

[10]

Q14.

(a) $\mathbf{v} = (3\mathbf{i} - 10\mathbf{j}) + (4\mathbf{i} + 2\mathbf{j})t$
Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= (3 + 4t)\mathbf{i} + (2t - 10)\mathbf{j}$
ag Correct result from correct working

A1

2

(b) $2t - 10 = 0$
j component equal to zero

M1

$t = 5$
Correct time

A1

2

(c) (i) $\mathbf{r} = (3\mathbf{i} - 10\mathbf{j}) \times 10 + \frac{1}{2} (4\mathbf{i} + 2\mathbf{j}) \times 10^2$
Finding $\mathbf{r}$ when $t = 10$

M1

Correct expression

A1

$= 230\mathbf{i}$
Correct final answer

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A1

3

(c) (ii) $\mathbf{v} = (3 + 4 \times 10)\mathbf{i} + (2 \times 10 - 10)\mathbf{j}$
 $= 43\mathbf{i} + 10\mathbf{j}$
Correct velocity

B1

1

(d) $\mathbf{r} = 230\mathbf{i} + (43\mathbf{i} + 10\mathbf{j}) \times 10$
Uses zero acceleration

M1

Uses both answers from (c)

M1

$= 660\mathbf{i} + 100\mathbf{j}$

Correct expression for r

A1

Correct simplified result

A1

$$r = \sqrt{660^2 + 100^2}$$

Finding magnitude

M1

$$= 668 \text{ m}$$

Correct distance

Follow through from part (c)

A1 ✓

6

[14]

Q15.

(a) $0^2 = 7^2 + 2(-9.8)s$

Use of constant acceleration equation with
 $v = 0$

M1

$$s = \frac{4.9}{19.6} = 2.5$$

Correct s

A1

Max Height = $5 + 2.5 = 7.5 \text{ m}$

ag Adding 5 to get total height

B1

3

(b) $0 = 7 - 9.8t$

Use of constant acceleration equation with $0 = v$

M1

$$t = \frac{7}{9.8} = 0.714 \text{ s}$$

ag Correct time from correct working

A1

2

- (c) (i) $5.5 = 4.9t^2$
Finding time to fall 5.5 metres

M1

$$t = \sqrt{\frac{5.5}{4.9}} = 1.059$$

Correct time

A1

$$\text{Total Time} = 0.7143 + 1.059 = 1.77 \text{ s}$$

Finding total time

m1

Correct time

A1

4

- (c) (ii) $v^2 = 0^2 + 2 \times 9.8 \times 5.5$
Use of constant acceleration based on catch at the correct height

M1

Correct equation

A1

$$v = \sqrt{107.8} = 10.4 \text{ ms}^{-1}$$

Correct speed

A1

3

[12]

Q16.

- (a) $R = 20 \times 9.8 = 196 \text{ N}$
Finding R and using $F = \mu R$

M1

$$F = 0.15 \times 196 = 29.4 \text{ N}$$

Correct friction

A1

2

- (b) $T - 29.4 = 20a$
Equation of motion for A

			M1	
	<i>Correct equation</i>			
			A1	
	$5 \times 9.8 - T = 5a$			
	<i>Equation of motion for B</i>			
			M1	
	<i>Correct equation</i>			
			A1	
	$49 - (20a + 29.4) = 5a$			
	$a = \frac{19.6}{25} = 0.784 \text{ ms}^{-2}$			
	ag <i>Correct acceleration from correct working</i>			
			A1	5
(c)	$T = 20 \times 0.784 + 29.4 = 45.1 \text{ N}$			
	<i>Substituting a into one equation of motion</i>			
			M1	
	<i>Correct tension</i>			
			A1	2
(d)	$s = \frac{1}{2} \times 0.784 \times 3^2$			
	<i>Use of constant acceleration equation with $u = 0$</i>			
			M1	
	$= 3.53 \text{ m}$			
	<i>Correct equation</i>			
			A1	
	<i>Correct distance</i>			
			A1	3

[12]

Q17.

- (a) $8 = 10 \cos 60^\circ t$
Horizontal equation based on travelling 8 metres

M1

Correct equation

A1

$$t = \frac{8}{10 \cos 60^\circ} = 1.6 \text{ s}$$

ag *Correct time from correct working*

A1

3

(b) $h = 10 \sin 60^\circ \times 1.6 - 4.9 \times 1.6^2$

Expression for height using $t = 1.6$

M1

Correct expression

A1

Correct height

A1

3

(c) $V = 10 \cos 60^\circ = 5$

Horizontal component of the velocity

B1

$$V = 10 \sin 60^\circ - 9.8 \times 1.6 = -7.020$$

Expression for vertical component of velocity

M1

Correct vertical component

A1

Finding the magnitude

M1

$$v = \sqrt{5^2 + 7.020^2} = 8.62 \text{ ms}^{-1}$$

Correct magnitude

A1

5

[11]

Q18.

(a) $5\mathbf{i} - 2\mathbf{j} = 4\mathbf{i} + 3\mathbf{j} + 10\mathbf{a}$

Use of constant acceleration equation for velocity

M1

Correct equation

A1

$$a = \frac{1}{10} (\mathbf{i} - 5\mathbf{j}) = (0.1\mathbf{i} - 0.5\mathbf{j}) \text{ ms}^{-2}$$

ag *Correct acceleration from correct working*

A1

3

(b) $\mathbf{r} = (4\mathbf{i} + 3\mathbf{j})t + 0.5(0.1\mathbf{i} - 0.5\mathbf{j})t^2$

Use of constant acceleration equation for position

M1

Correct expression

A1

2

(c) $\mathbf{r} = (4t + 0.05t^2)\mathbf{i} + (3t - 0.25t^2)\mathbf{j}$

j component is zero

M1

Correct equation

A1

$$3t - 0.25t^2 = 0$$

$$t(3 - 0.25t) = 0$$

Solving for t

$$t = 0 \text{ or } t = \frac{3}{0.25} = 12 \text{ s}$$

$$t = 12 \text{ s}$$

Correct time

A1

4

(d) $\mathbf{v} = (4 + 0.1t)\mathbf{i} + (3 - 0.5t)\mathbf{j}$

$$3 - 0.5t = 0$$

$$t = 6$$

Expression for velocity

M1

Correct velocity

A1

j component is zero

m1

Correct time

A1

4

[13]

Q19.

(a) (i) $6^2 = 2^2 + 2 \times a \times 10$

Use of a constant acceleration equation to find a

M1

$$a = \frac{36 - 4}{20} = 1.6 \text{ ms}^{-2}$$

Correct result from correct working

A1

2

(ii) $6 = 2 + 1.6t$

Use of a constant acceleration equation to find t

M1

$$t = \frac{4}{1.6} = 2.5 \text{ s}$$

Correct t from correct working

A1

2

(b) $F - 35 = 65 \times 1.6$

$$F = 104 + 35 = 139 \text{ N}$$

Three term equation of motion

M1

Correct equation

A1

Correct force

A1

Q20.

(a) $60\mathbf{i} + 20\mathbf{j} = 20(2\mathbf{i} - 3\mathbf{j}) + 200\mathbf{a}$

$$20\mathbf{i} + 80\mathbf{j} = 200\mathbf{a}$$

$$\mathbf{a} = 0.1\mathbf{i} + 0.4\mathbf{j}$$

*Use of constant acceleration equation
in vector form to find $\mathbf{a}$*

M1

Correct equation

A1

Correct $\mathbf{a}$

A1

3

(b) $\mathbf{v} = (2\mathbf{i} - 3\mathbf{j}) + (0.1\mathbf{i} + 0.4\mathbf{j})t$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$$= (2 + 0.1t)\mathbf{i} + (-3 + 0.4t)\mathbf{j}$$

Correct expression

A1

2

(c) $2 + 0.1t = -(-3 + 0.4t)$

Equating components with $\pm$

M1

$$0.5t = 1$$

Correct equation

A1

$$t = 2$$

A1

$$\mathbf{v} = 2.2\mathbf{i} - 2.2\mathbf{j}$$

Finding velocity

M1

$$v = \sqrt{2.2^2 + 2.2^2} = 3.11 \text{ ms}^{-1}$$

Correct speed

A1

5

[10]

Q21.

(a) $9 = 3 + 1.2t$

Forming constant acceleration equation

M1

$$t = \frac{9 - 3}{1.2} = 5 \text{ seconds}$$

Correct result from correct working

A1

2

(b) $s = \frac{1}{2} (3 + 9) \times 5 = 30 \text{ metres}$

Forming constant acceleration equation

Correct distance

M1 A1

2

(c) $F = 1200 \times 1.2 = 1440 \text{ N}$

Applying Newton's second law with $a = 1.2$

Correct F

M1 A1

2

[6]

Q22.



B1

1

(b) $R = 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$

Correct force diagram

Resolving perpendicular to slope

Correct R

M1 A1

2

(c) $F = 0.2R = 7.51\text{N}$

Using $F = \mu R$

Correct F from correct working

M1 A1

2

(d) $5 \times 9.8 \sin 40^\circ - F = 5a$

Resolving parallel to slope to give 3 term equation of motion

M1 A1

Correct equation

$$a = \frac{5 \times 9.8 \sin 40^\circ - F}{5} = 4.80 \text{ ms}^{-2}$$

m1

Solving for a

Correct a from corret working

A1

4

(e) $10^2 = 2^2 + 2 \times 4.80s$

Forming constant acceleration equation

Correct equation

M1 A1

$$s = \frac{100 - 4}{9.6} = 10.0 \text{ m}$$

Correct s

A1

3

[12]

Q23.

(a) $19\mathbf{i} - 25\mathbf{j} = \frac{1}{2} \mathbf{a} \times 10^2 + 9\mathbf{i} + 10\mathbf{j}$

Using both position vectors to form a constant acceleration equation

M1

$$50\mathbf{a} = 10\mathbf{i} - 35\mathbf{j}$$

Correct equation

Solving for a

A1 M1

$$\mathbf{a} = 0.2\mathbf{i} - 0.7\mathbf{j}$$

Correct a

A1

4

(b) $\mathbf{v} = 10(0.2\mathbf{i} - 0.7\mathbf{j})$

use of $\mathbf{v} = \mathbf{at}$

M1

$$= 2\mathbf{i} - 7\mathbf{j}$$

Correct velocity

A1

$$v = \sqrt{2^2 + 7^2} = 7.28 \text{ ms}^{-1}$$

Finding speed from velocity

Correct speed

m1 A1

4

(c) $15.4 = \frac{1}{2} \times 0.2 \times t^2 + 9$

Finding t from one component

M1

$$t = 8$$

Correct t

A1

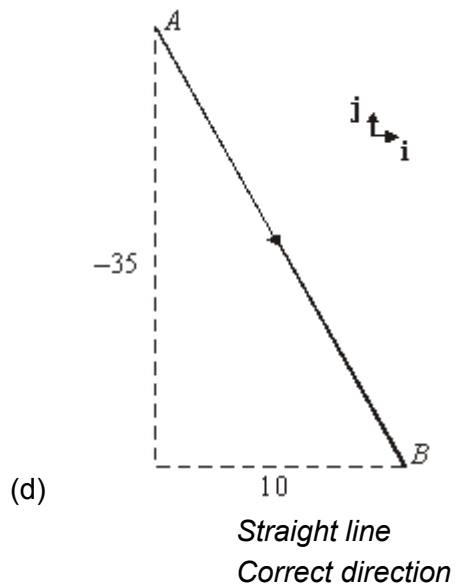
$$\frac{1}{2} \times (-0.7) \times 8^2 + 10 = -12.4$$

Using $t = 8$ with other component

Correct result

M1 A

4



B1 B1

2

[14]

Q24.

(a) (i) $s_1 = \frac{1}{2} \times 0.2 \times 8^2 = 6.4 \text{ m}$
Constant acceleration equation with $u = 0$
Correct distance

M1A1

2

(ii) $v = 0.2 \times 8 = 1.6 \text{ ms}^{-1}$
Constant acceleration equation with $u = 0$

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M1

2

(iii) $s_2 = 1.6 \times 3 = 4.8 \text{ m}$

M1

$s = 6.4 + 4.8 + 1 = 12.2 \text{ m}$
Finding stage II distance
Correct total distance

A1

2

(b) $1 = \frac{1}{2} (0 + 1.6)t_3$
Equation to find stage III time

M1

$$t_3 = 1.25$$

Correct time

A1

$$t = 8 + 3 + 1.25 = 12.25 \text{ s}$$

Correct total time

A1

3

(c) $T - 600 \times 9.8 = 600 \times 0.2$

Three term equation of motion

Correct equation

M1 A1

$$T = 6000 \text{ N}$$

Correct tension

A1

3

[12]

Q25.

(a) $40\mathbf{i} - 15\mathbf{j} = \frac{1}{2} (5\mathbf{i} - 2\mathbf{j} + \mathbf{v}) \times 10$

Use of constant acceleration equation with $\mathbf{v}$ unknown

Correct equation

M1 A1

$$\mathbf{v} = 8\mathbf{i} - 3\mathbf{j} - 5\mathbf{i} + 2\mathbf{j} = 3\mathbf{i} - \mathbf{j}$$

Solving for $\mathbf{v}$

Correct $\mathbf{v}$

M1 A1

4

(b) $3\mathbf{i} - \mathbf{j} = 5\mathbf{i} - 2\mathbf{j} + 10\mathbf{a}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1A1

$$\mathbf{a} = -0.2\mathbf{i} + 0.1\mathbf{j}$$

Correct $\mathbf{a}$

A1

3

(c) $\mathbf{F} = 15(-0.2\mathbf{i} + 0.1\mathbf{j})$

Use of $\mathbf{F} = m\mathbf{a}$

M1

$= -3\mathbf{i} + 1.5\mathbf{j}$

Correct $\mathbf{F}$

A1

$F = \sqrt{3^2 + 1.5^2} = 3.35$

Finding magnitude of $\mathbf{F}$

Correct magnitude

M1 A1

4

[11]

Q26.

(a) $24.5 = 9.8t$

Use of $v = u + at$ with $u = 0$

M1

$t = \frac{24.5}{9.8} = 2.5 \text{ seconds}$

Correct time

A1

2

(b) $24.5^2 = 0^2 + 2 \times 9.8s$

*Use of constant acceleration equation to find s ,
with $u = 0$ or $v = 24.5$*

Correct equation

M1

$s = \frac{24.5^2}{2 \times 9.8} = 30.625 \text{ m}$

ag *Correct distance from correct working*

A1

3

(c) $30.625 - 5 = 4.9t^2$

Use of $s = ut + \frac{1}{2}at^2$ with $u = 0$

M1

Correct equation

A1

$$t = \sqrt{\frac{25.625}{4.9}} = 2.29$$

Solving for t having subtracted 5

m1

4

Correct t

A1

[9]

Q27.

(a) $R = 20 \times 9.8 = 196$

cao

B1

1

(b) $F \leq 0.3 \times 196 = 58.8$

Using 0.3×196

M1

If $P = 80$, $F = 58.8$

58.8 as answer

A1

If $P = 40$, $F = 40$

40 as answer

A1

3

(c) $P - 58.8 = 20 \times 0.8$

Three term equation of motion including 58.8

M1

Correct equation

A1

$P = 74.8$

A1

Correct P

A1

3

(d) $a = \frac{-58.8}{20} = -2.94$

Use of $F = ma$ with ± 58.8

M1

Correct acceleration with a negative sign

A1

$$0^2 = 6^2 + 2 \times (-2.94)s$$

Use of $v^2 = u^2 + 2as$ with $v = 0$

m1 A1

$$s = \frac{36}{5.88} = 6.12$$

Correct distance

A1

5

[12]

Q28.

(a) $0.5 = \frac{1}{2} \times a \times 4$

Use of $s = ut + \frac{1}{2}at^2$ with $u = 0$

M1

$$a = 0.25$$

Correct acceleration

A1

2

(b) $6 \times 9.8 - T = 6 \times 0.25$

Three term equation of motion for particle,
with correct use of g

M1

Correct equation

A1ft

$$T = 57.3$$

ag Correct T from correct working

A1ft

3

(c) $57.3 - F = 10 \times 0.25$

Three term equation of motion for the block

M1

Correct equation

A1ft

$$F = 54.8$$

Correct F

A1ft

$$R = 10 \times 9.8 = 98$$

R = 98 seen in working

B1

$$54.8 = 98\mu$$

Use of $F = \mu R$

m1

$$\mu = \frac{54.8}{98} = 0.559$$

correct μ

A1ft

6

[11]

Q29.

(a) (i) $\mathbf{a} = \frac{1}{4} (8\mathbf{i} - 3\mathbf{j}) = 2\mathbf{i} - 3\mathbf{j}$

Use of $\mathbf{F} = m\mathbf{a}$, must be applied to both components

M1

Correct acceleration

A1

2

(ii) $\mathbf{v} = 20(2\mathbf{i} - 3\mathbf{j}) = 40\mathbf{i} - 60\mathbf{j}$

Use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$ with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

		M1	
	Correct v		
		A1	
			2
(iii)	$r = \frac{1}{2} (2i - 3j) \times 20^2$ Use of constant acceleration equation to find r with $u = 0i + 0j$		
		M1	
	Correct expression		
		A1	
	$= 400i - 600j$ Correct r in simplified form		
		A1	
			3
(b)	$r = 400i - 600j + 25(40i - 60j)$ Use of $r + 25v$		
		M1	
	Correct expression		
		A1ft	
	$= 1400i - 2100j$ Correct position vector		
		A1ft	
	$r = \sqrt{1400^2 + 2100^2} = 2520 \text{ m (to 3 sf)}$ Finding magnitude		
		m1	
	Correct distance from correct working		
		A1ft	
			5
	Alternative: Straight line method M1 two distances r and $25v$ A1 for each distance m1 adding A1 correct final answer		

[12]

Q30.

(a) $5 = 32\sin 60^\circ t - 4.9t^2$

M1

Equation for vertical motion with ± 5

M1

LHS correct

A1

RHS correct

A1

$$4.9t^2 - 32\sin 60^\circ t + 5 = 0$$

$$t = 0.1866 \text{ or } 5.4691$$

Solving quadratic

m1

5.47 seconds

Selecting larger answer from two solutions or obtaining one answer with a reason

A1

5

(b) $32\cos 60^\circ \times 5.469 = 87.5 \text{ m}$

Equation for horizontal motion

M1

Correct range

A1

2

(c) $v_H = 32\cos 60^\circ$

Horiz. component of velocity seen or used

B1

$$v_V = 32\sin 60^\circ - 9.8 \times 5.469 = -25.88$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{16^2 + 25.88^2} = 30.4 \text{ ms}^{-1}$$

Finding magnitude

m1

Correct speed
(Note Max Height = 39.2 m from $t = 2.83$)

A1

5

[12]

Q31.

(a) $v = 0 + 0.5 \times 10 = 5 \text{ ms}^{-1}$

Use of $v = u + at$

M1

Correct v

A1

2

(b) $s = \frac{1}{2} \times 0.5 \times 10^2 = 25 \text{ m}$

Use of a constant acceleration equation to find s

M1

Correct s

A1

2

(c) $F - 40000 = 200000 \times 0.5$

Three term equation of motion

M1

Correct equation

A1

$F = 140000 \text{ N}$

Correct force

A1

3

[7]

Q32.

(a) (i) $16^2 = 0^2 + 2 \times a \times 400$

Using constant acceleration equation to find a

M1

$$a = \frac{256}{800} = 0.32 \text{ ms}^{-2}$$

ag Correct a from correct working

A1

2

(ii) $16 = 0 + 0.32t$

Using constant acceleration equation to find t

M1

$$t = \frac{16}{0.32} = 50 \text{ seconds}$$

Correct t

A1

2

(b) (i) $30^2 = 16^2 + 2 \times 0.5s$

Using constant acceleration equation to find s

M1

$$s = 30^2 - 16^2 = 644\text{m}$$

Correct s

A1

2

(ii) $644 = \frac{1}{2}(16 + 30)t$

Using constant acceleration equation to find t

M1

$$t = \frac{644}{23} = 28\text{s}$$

Correct t

A1

$$\text{total time} = 28 + 50 = 78 \text{ s}$$

Adding 50 to give total time

A1f.t.

3

[9]

Q33.

(a) $12g - T = 12a$

Equation of motion for one particle

M1

Correct equation

A1

$T - 8g = 8a$

Equation of motion for other particle

M1

Correct equation

A1

$8a + 8g = 12g - 12a$

$20a = 4g$

$a = \frac{4g}{20} = 1.96 \text{ ms}^{-2}$

ag *Correct a from correct working*

A1

5

(b) $T = 8 \times 1.96 + 8 \times 9.8$

$= 94.1 \text{ N}$

Substituting value for a into equation of motion to find T

M1

Correct T

A1

2

(c) $7 = 0 + 1.96t$

Using constant acceleration to find t

M1

$t = \frac{7}{1.96} = 3.57 \text{ s (to 3 sf)}$

Correct t

A1

2

[9]

Q34.

(a) $19\mathbf{i} + 13\mathbf{j} = 35\mathbf{i} + 45\mathbf{j} + 8\mathbf{a}$

Constant acceleration equation to find a

M1

$$\mathbf{a} = \frac{19-35}{8}\mathbf{i} + \frac{13-45}{8}\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$$

Correct equation

A1

ag *Correct a from correct working*

A1

3

(b) $\mathbf{r} = (35\mathbf{i} + 45\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} - 4\mathbf{j})t^2$

Use of constant acceleration equation

M1

Correct i component

A1

Correct j component

A1

3

(c) $\mathbf{r} = (3.5t - t^2)\mathbf{i} + (45t - 2t^2)\mathbf{j}$

Splitting into components correctly

B1

$$35t - t^2 = 300$$

Forming equation for one component

M1

$$t^2 - 35t + 300 = 0$$

Solving quadratic

M1

$$t = 15 \text{ or } t = 20$$

Two correct solutions

A1

$$4t - 2t^2 = 225$$

Forming correct second quadratic

B1

$$2t^2 - 45t + 225 = 0$$

$$t = 7.5 \text{ or } t = 15$$

Solving quadratic for two solutions

M1

$$t = 15 \text{ seconds}$$

Correct final solution

A1

7

[13]

Q35.

(a) $s = \frac{1}{2} (40 + 60) \times 4$

Equation based on area of trapezium of rectangle plus two triangles

Correct equation

M1

A1

$$= 200\text{m}$$

Correct final distance

A1

3

(b) $a = \frac{4}{5} = 0.8 \text{ ms}^{-2}$

Attempting to find acceleration

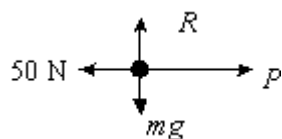
M1

Correct acceleration

A1

2

(c) (i)



Correct force diagram

B1

1

(ii) $P - 50 = 65 \times 0.8$

Three term equation of motion

M1

Correct equation

A1

$P = 102 \text{ N}$

Correct value

A1

3

[9]



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